Project

0 How does vaccinating a population impact the spread of disease?

1 Underlying Processes

- You can catch covid if: no symptoms, vaccinated, had covid before
- Most people aren't infectious after 5 days but some up to 10
- Immunity last 5-8 months

2 Compartments

Susceptable
Infected or Asymptomatic
Recovered

3 Assumptions

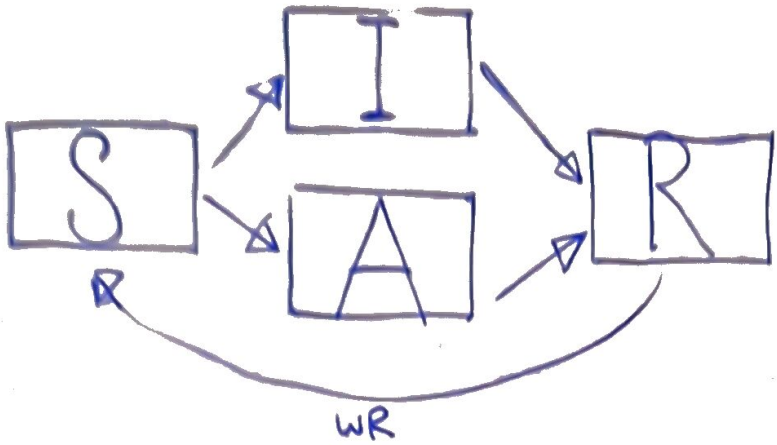
1) For simplicity, Infected compartment includes those who are pre-symptomatic
2) No longer infected after 7days
3) Immunity lasts 6 months
4) 20% will become asymptomatic
5) Dead included in recovered

$\beta = 3$ ← indiv transmission Rate
new cases per time unit

$\frac{\text{Quantity}}{\text{time}}$

New Infections $= \beta \times I \times \frac{S}{\text{pop size}}$. life span $= 80 \times 365$

recoveries per day $= \gamma I = \frac{I}{\text{life span}}$



$\beta = 3$ → # new cases per day

$$\Delta S = \left(\frac{\beta I S}{N} + \frac{\beta A S}{N} + wR\right)$$

$$\Delta A = 0.2\left(\frac{\beta I S}{N} + \frac{\beta A S}{N} + wR\right) - \gamma_A A$$

asymptomatic

$$\Delta I = 1 - 0.2\left(\frac{\beta I S}{N} + \frac{\beta A S}{N} + wR\right) - \gamma I$$

$$\Delta R = \gamma_A A + \gamma I - wR$$

$$wR = 1 - 0.06(\gamma_A A + \gamma I)$$

mortality rate